FEAT: A Preference Feedback Dataset through a Cost-Effective Auto-Generation and Labeling Framework for English AI Tutoring

Anonymous ACL submission

Abstract

In English education tutoring, teacher feedback is essential for guiding students. Recently, AI-based tutoring systems have emerged to assist teachers; however, these systems require high-quality and large-scale teacher feedback data, which is both time-consuming and costly to generate manually. In this study, we propose FEAT, a cost-effective framework for generating teacher feedback, and have constructed three complementary datasets: (1) DIRECT-Manual (DM), where both humans and large language models (LLMs) collaboratively generate high-quality teacher feedback, albeit at a higher cost; (2) DIRECT-Generated (DG), an LLM-only generated, cost-effective dataset with lower quality;, and (3) DIRECT-Augmented (DA), primarily based on DG with a small portion of DM added to enhance quality while maintaining cost-efficiency. Experimental results showed that incorporating a small portion of DM (5–10%) into DG leads to superior performance compared to using 100% DM alone.

1 Introduction

In English education tutoring, providing appropriate teacher feedback plays a crucial role in guiding students and improving their educational outcomes (Ma et al., 2014; Fossati, 2008). Given its importance, various studies have explored automated teacher feedback generation (Meyer et al., 2024; Scarlatos et al., 2024b; Liu et al., 2020).

Figure 1 illustrates methods for generating and annotating teacher feedback for tutoring systems. As shown in (a), human-generated feedback with ranking provides high quality, but its time-consuming and costly nature makes it difficult to scale up (Chang et al., 2023).

To address this challenge, we propose **FEAT**, a cost-effective framework using large language models (LLMs) to automatically generate a large-scale teacher feedback preference dataset for tutoring AI.

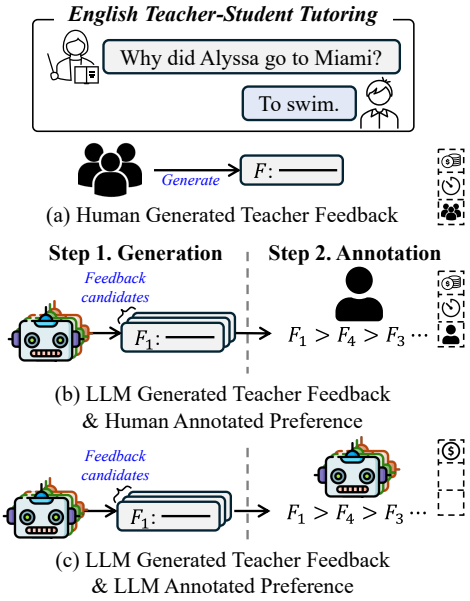


Figure 1: Teacher feedback generation and annotation process in an English tutoring system.

This enables reward- or rank-based learning, making it suitable for building human-friendly tutoring models (Ouyang et al., 2022). FEAT generates teacher feedback based on student responses, using the dialogue history between teacher and student and context as input. Moreover, we apply feedback criteria defined by Seo et al. (2025) to ensure educationally appropriate feedback.

Using FEAT, we constructed three datasets: (1) *DIRECT-Manual* (DM), which contains human and LLM-generated feedback with human-annotated rankings (high quality and high cost), (2) *DIRECT-Generated* (DG), an entirely LLM-generated and annotated preference dataset (medium quality and low cost), and (3) *DIRECT-Augmented* (DA), a hybrid dataset built on DG with a minor addition of DM (high quality and low cost)[1].

Our experiments showed that incorporating a small portion of DM (5–10%) into DG leads to superior performance compared to using DM alone.

[1]We will release FEAT and its datasets upon publication.

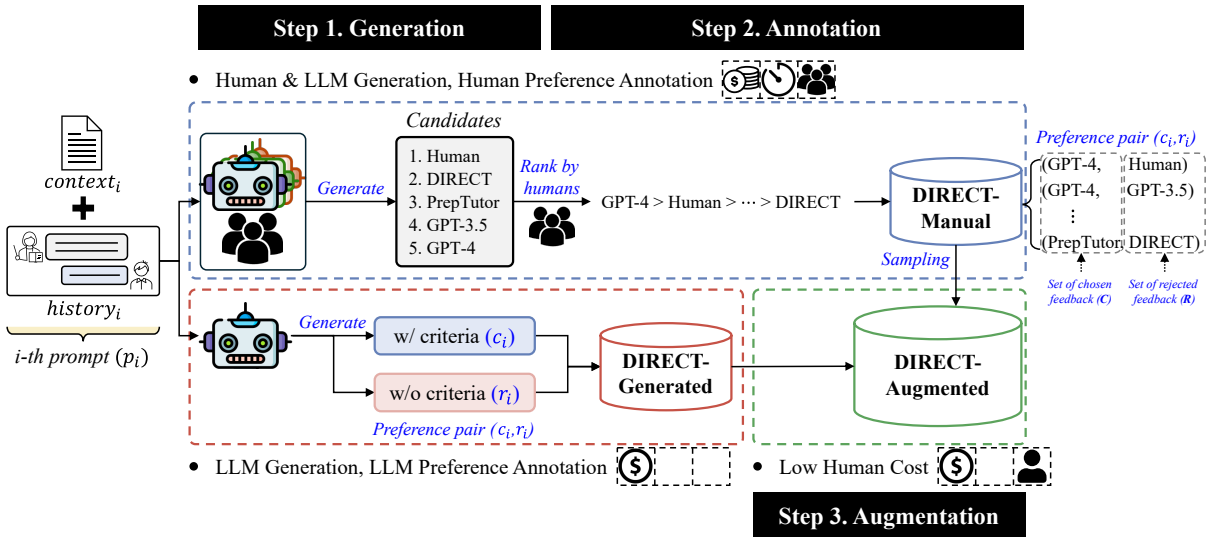


Figure 2: The architecture of the FEAT framework, illustrating the construction process of the DIRECT-Manual, DIRECT-Generated, and DIRECT-Augmented datasets. p_i, c_i, and r_i denote the i-th prompt, chosen, and rejected responses, respectively.

Our main contributions are as follows:

- We proposed FEAT, a cost-effective framework for automated teacher feedback generation and annotation in English tutoring.
- We constructed three preference datasets: DM, DG, and DA, enabling reward- or rank-based learning.
- We confirmed that incorporating a small amount of DM into DG (DA) yielded better performance than DM alone.

2 FEAT: *F*eedback Dataset Generation Framework for *E*nglish *A*I *T*utoring

Figure 2 illustrates the construction process of our FEAT framework. FEAT applies five criteria from Seo et al. (2025)-*Correct, Revealing, Guidance, Diagnostic,* and *Encouragement*—ensuring educationally effective feedback.

2.1 DIRECT-Manual: Rank-based Preference Dataset

DM is an extended version of the DIRECT (Huang et al., 2023) dataset, simulating intelligent tutoring between teachers and students. While it ensures high quality, it relies heavily on human effort, making it time-consuming and costly.

Step 1: Feedback Generation. We collected teacher feedback data for scenarios with incorrect student answers in teacher-student dialogues from diverse sources (Human, DIRECT, PrepTutor, GPT-3.5, and GPT-4; see Appendix B), with DM previously used as private data in Liermann et al. (2024).

Step 2: Feedback Ranking via Human Annotation. Human annotators ranked the five feedback candidates using two criteria: *Correct* (specific and accurate information) and *Revealing* (avoiding direct answers). Feedback meeting both criteria received the highest rank, with Correct prioritized when only one criterion was met.

Step 3: Prefernce Data Construction. From ranked five feedback candidates, we created pairwise combinations; in each pair, the higher-ranked feedback is labeled ***chosen*** and the lower-ranked as ***rejected***.

2.2 DIRECT-Generated: Criteria-based Preference Dataset

DG uses LLM to automatically generate and annotate teacher feedback based on specific criteria, producing reasonably good data at a lower cost.

Step 1: Feedback Generation. Using dialogue history and context, LLM generates teacher feedback based on five criteria. We created tutoring scenarios by converting reading comprehension tasks from MCTest (Richardson et al., 2013) to generate large-scale feedback data.

Step 2: Preference Data Construction. We generated two types of feedback: *w/ criteria* (applying five criteria) and *w/o criteria* (without criteria). We labeled *w/ criteria* feedback as ***chosen*** and *w/o criteria* as ***rejected***, assuming criteria-based feedback is of higher quality. Data statistics are shown in Table 1, with full details in Appendix C.

3 Teacher Feedback Ranking

To validate the preference annotations in DM, DG, and DA, we trained pairwise-based ranking models.

3.1 Ranking Models

We employed five approaches to train ranking models. Each model takes *(prompt, chosen, rejected)* as input.

Binary Classifier formulates preference learning as a binary classification task, labeling *(chosen, rejected)* pairs as 1 and *(rejected, chosen)* pairs as 0. The input sequence is depicted in Figure 11.

Reward Model (Ouyang et al., 2022) computes scalar preference scores for feedback pairs, training to assign higher scores to chosen feedback.

Direct Preference Optimization (DPO) (Rafailov et al., 2023) optimizes language model probabilities to prefer chosen feedback, using log probability differences between chosen and rejected pairs.

RankNet (Burges et al., 2005) learns score differences between feedback pairs using Binary Cross-Entropy loss. Reward Model, DPO, and RankNet share the same prompt, shown in Figure 12.

Ensemble aggregates predictions from the above four approaches through majority voting.

3.2 Scenarios for Training

We evaluated our ranking models on DM with three training configurations. The arrow (→) indicates training (left) and evaluation (right).

- **DM→DM**: Training with DM using manual annotation, serving as a performance upper bound for comparison with the other two scenarios (DG→DM and DA→DM).
- **DG→DM**: Training with DG using automatic annotation.
- **DA→DM**: Hybrid training using DG combined with a subset of DM for mixed annotation.

During training, we enhanced data diversity by including feedback from different contexts beyond the standard *(chosen, rejected)* pairs. This approach enabled the model to learn feedback comparisons across various contexts.

For evaluation, we tested the model on all possible pairs in DM. The model's pairwise predictions were aggregated to create overall rankings, with accuracy scored as 1 for *chosen* > *rejected* and 0 for *chosen* < *rejected*.

	Train	Test
DIRECT-M	5,025	475
DIRECT-G	3,996	444

Table 1: Dataset statistics.

4 Experiments

We designed experiments and analyzed results to address the following research questions:

- How does ranking model performance with DG compare to human-curated DM (Section 4.2)?
- How does the ratio of DM in DA affect ranking model performance (Section 4.3)?
- How does the number of criteria in DG affect ranking models performance? (Section 4.4)?

4.1 Experimental Setup

Models We trained ranking models using five open-source models: Llama-1B (Dubey et al., 2024), Llama-1B-IT, Llama-3B, Llama-3B-IT, and Qwen-3B-IT (Bai et al., 2023). Model details and hyperparameters are provided in Appendix D.

Evaluation Metrics Rank-biased overlap (RBO) scores range from 0 to 1, where 1 indicates perfect ranking similarity.

4.2 Comparison of Ranking Model Performance

As shown in Figure 3, DM→DM performed consistently (0.77-0.80) across all sizes and approaches. While DG→DM showed lower but competitive results: the binary classifier reached 0.76 (Llama-1B), reward model 0.73 (Llama-3B-IT), DPO 0.71 (Llama-3B-IT, Qwen-3B-IT), RankNet 0.76 (Qwen-3B-IT), and Ensemble 0.76 (Qwen-3B-IT). Notably, the Ensemble maintained stable performance across different architectures, mitigating the variability seen in individual approaches.

These results indicated that teacher feedback generated by LLMs can produce rankings highly comparable to human annotator rankings, with a particularly strong trend observed in larger models. Full results are in Table 7 (Appendix E).

4.3 Performance Analysis by DM Ratio in the DA→DM Scenario

We analyzed how varying the proportion of DM (5-100%) in the DA→DM scenario affects model per-

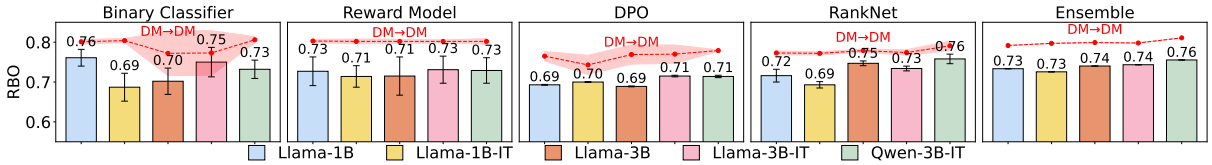
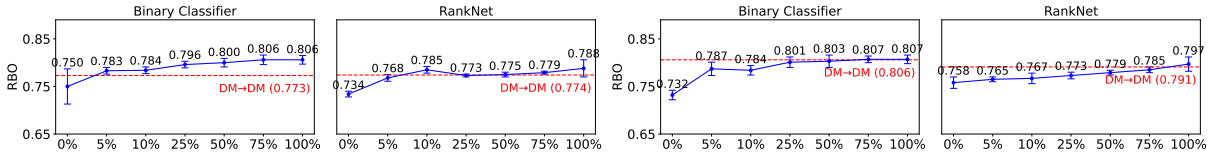


Figure 3: Ranking model performance across different approaches (with 5-seed standard deviation). Lines indicate DM→DM performance, while bars show DG→DM performance.



(a) Llama-3B-IT (b) Qwen-3B-IT

Figure 4: Binary Classifier and RankNet performance in the DA→DM scenario. (See Appendix E for other approaches.)

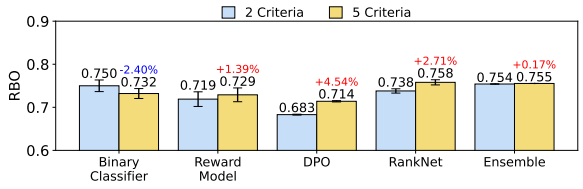


Figure 5: Performance of the Qwen-3B-IT across varying numbers of feedback criteria. See Appendix E for additional results.

formance. This experiment was conducted on the Llama-3B-IT and Qwen-3B-IT only, which demonstrated higher performance in most settings. Figure 4 presents the performance of the binary classifier and RankNet, which achieved the highest results in Section 4.2.

For Llama-3B-IT, the binary classifier outperformed the DM→DM even only 5% of human-annotated DM and DA. Similarly, RankNet surpassed DM→DM performance in the 5–10% range. In contrast, Qwen-3B-IT exceeded DM→DM performance in most cases, particularly in the 75–100% range. Although Qwen-3B-IT is not as efficient as Llama-3B-IT, the results suggest that high performance can be achieved with minimal human cost. The overall performance of all approaches is shown in Figure 13 (see Appendix E).

4.4 Performance Analysis by Number of Feedback Criteria

To investigate the impact of feedback criteria, we compared two training configurations: using all five criteria versus using only two essential criteria (Correct and Revealing). For the experiments, we generated an additional version of DG that includes only two feedback criteria and trained a ranking model in the DG→DM scenario.

Figure 5 shows that Qwen-3B-IT models trained with five criteria demonstrated improvements across most approaches, with DPO and RankNet achieving notable gains (+4.54% and +2.71% respectively). This suggests that incorporating richer feedback information enhances model generalization, although the binary classifier showed no benefit from additional criteria. Full results for all models are available in Figure 14 (Appendix E).

5 Conclusion

In this study, we proposed the *F*eedback Dataset Generation Framework for *E*nglish *AI* *T*utoring (FEAT), which utilizes LLMs to generate teacher feedback and build preference datasets for English tutoring. We evaluated ranking models on three datasets—DIRECT-Manual (DM), DIRECT-Generated (DG), and DIRECT-Augmented (DA)—constructed via FEAT.

Results showed that models based on DG performed competitively with DM-based models. Moreover, supplementing with only 5–10% human-annotated DM led to superior performance than using the full DM dataset. These findings demonstrate that high performance can be achieved with minimal human effort with our FEAT framework. In future research, we will extend our framework to broader educational scenarios.

Limitations

In this study, we explored the feasibility of LLM-based teacher feedback generation and preference dataset construction using the FEAT framework. However, the study has the following limitations:

First, while we constructed an English tutoring scenario using the MCTest dataset, further research is required to assess the generalizability of the framework across diverse educational datasets.

Second, we only conducted the ranking model experiments using 1B and 3B LLMs. Future work should explore the applicability of larger LLMs (e.g., 7B, 13B, 70B) to evaluate their impact on ranking performance.

Third, we employed a pairwise approach for ranking model training. We plan to explore preference dataset construction and training strategies applicable to pointwise and listwise ranking approaches in future research.

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A Related Works

A.1 Feedback in Education

In the field of education, teacher feedback plays a crucial role in enhancing students' learning experiences and achievements. In particular, immediate and appropriate feedback positively impacts students' cognitive, emotional, and motivational outcomes (Shute, 2008).

Research has been conducted on designing effective feedback strategies. Nicol and Macfarlane-Dick (2006) proposed seven principles for effective feedback, while Hattie and Timperley (2007) analyzed the impact of feedback on learning and investigated its key components. Steiss et al. (2024); Scarlatos et al. (2024c) proposed five criteria for evaluating feedback quality, designing them to help students understand clear directions for improvement and maintain motivation. Additionally, research applying feedback criteria has been conducted in fields such as programming education (Koutcheme et al., 2024).

A.2 Large Language Models in Education

Advancements in large language models (LLMs) have significantly impacted the field of education

(Gan et al., 2023; Wang et al., 2024; Lan and Chen, 2024; Jeon and Lee, 2023; Dai et al., 2023). The integration of LLMs with educational technology has been applied across various domains, including automated short answer grading (Kortemeyer, 2024), automated essay scoring (Stahl et al., 2024), automated distractor generation (Feng et al., 2024; Scarlatos et al., 2024a), and automatic question generation (Luo et al., 2024a; Lee et al., 2024; Ashok Kumar and Lan, 2024; Mulla and Gharpure, 2023; Wang et al., 2022).

Research has also been conducted on automated feedback systems to provide better feedback to students (Dai et al., 2023; Meyer et al., 2024). Additionally, LLM-powered personalized feedback generation has contributed to reducing teachers' workloads and improving the efficiency of online education (Liu et al., 2020). Beyond feedback generation, LLMs have also been utilized for feedback quality assessment. Studies have proposed LLM-based feedback generation and evaluation frameworks in domains such as programming assignments (Koutcheme et al., 2024) and mathematics (Scarlatos et al., 2024c), demonstrating the potential of LLMs in educational settings.

A.3 Learning to Rank Approaches

Learning to Rank (LTR) is widely used in information retrieval and recommendation systems, aiming to learn the optimal ranking of items for a given query. LTR methodologies are generally categorized into three approaches: Pointwise, Pairwise, and Listwise.

Pointwise approaches, such as MCRank (Li et al., 2007) and PRank (Crammer and Singer, 2001), predict a relevance score for each item individually and rank them based on these scores. Pairwise approaches learn the relative preference between two items, with representative algorithms including RankNet (Burges et al., 2005), LambdaRank (Burges et al., 2006), RankSVM (Joachims, 2002), RankBoost (Freund et al., 2003), GBRank (Zheng et al., 2007), and FRank (Tsai et al., 2007). Listwise approaches consider the entire item list as a single input and optimize its order holistically, with prominent algorithms such as ListNet (Cao et al., 2007), ListMLE (Lan et al., 2014), and SoftRank (Taylor et al., 2008).

Recent LTR research has expanded to leverage LLMs for ranking tasks. Qin et al. (2024); Luo et al. (2024b) proposed LLM-based pairwise ranking methods, demonstrating the potential of large-scale language models in ranking optimization.

	Human	DIRECT	PrepTutor	GPT-3.5	GPT-4
Word	10.02	9.27	29.37	17.84	21.01
Token	13.98	13.25	36.60	22.61	26.58

Table 2: Average feedback word length and token length in the DIRECT-Manual dataset.

	# Data	Average Words Per:	
		Story	Question
DIRECT-Manual	5,500	193.05	11.90
MCTest	1,480	202.71	7.79

Table 3: DIRECT-Manual and MCTest dataset statistics.

B Details of DIRECT-Manual Dataset

B.1 Feedback Generation Process

The DIRECT-Manual (DM) consists of five feedback candidates, each generated through different methods. The details of these candidates are as follows:

- **Human**: Feedback written by human annotators.
- **DIRECT**: Feedback generated using GPT-2 trained on the DIRECT dataset.
- **PrepTutor**: Feedback generated using GPT-2 fine-tuned on external domain-specific feedback data.
- **GPT-3.5**: Feedback generated using GPT-3.5-turbo-0613.
- **GPT-4**: Feedback generated using GPT-4-0613.

B.2 Feedback Ranking Process

The feedback candidates were ranked by human annotators based on two criteria:

- **Correct**: The feedback provides specific factual information based on the student's response or the given text.
- **Revealing**: The feedback guides the student toward the correct answer without explicitly stating it.

Table 2 presents the average length of feedback candidates, while Table 3 provides overall dataset

Story
Bill is ten years old.
One day his friend Tom says to him, “I am going to have a birthday party on Sunday evening. Bill, can you come to my party? ” “I’ll be glad to,” answers Bill.
...

History
Teacher : All right. Do you know how old Bill is?
Student : He is eight years old.

Reference
Ten. (Bill is ten years old.)

Feedback Candidates
- Human: Pretty close, but he is a bit older than that.
- DIRECT: no, he is one year older than that.
- PrepTutor: can you think more? honestly, bill is now 10 years old and he still has lots of growth. how old is he?
- GPT-3.5: Hmm, that's close. But is there any information in the passage that tells us Bill's age?
- GPT-4: Take a closer look at the beginning of the passage. It mentions Bill's age there.

Ranking
GPT-4>Human>GPT-3.5>PrepTutor>Tutor

Figure 6: Sample from the DIRECT-Manual.

statistics. An example from the DM is shown in Figure 6. Figure 7 illustrates the prompt used for feedback generation with GPT-3.5 and GPT-4 in the DM.

C Details of DIRECT-Generated Dataset

C.1 Dataset Preprocessing Process

The DIRECT-Generated (DG) dataset was constructed based on MCTest (Richardson et al., 2013), which consists of stories designed for students in grades 1–4, along with corresponding questions and four answer options.

The dataset construction involved the following preprocessing steps:

1. The question field from MCTest was used as the teacher’s question.
2. The answer field was used as the student’s correct response.
3. One option from the answer choices was randomly selected as the student’s incorrect response.

A sample from the MCTest is illustrated in Figure 8.

C.2 Feedback Generation Process

We utilized LLMs to automatically generate teacher feedback. The prompt for feedback generation was designed to include the story, question, the student’s incorrect response, and the correct response. Additionally, the five feedback criteria defined by Seo et al. (2025) were applied to ensure educationally effective feedback. The characteristics of each criterion are as follows:

Prompt for generating DIRECT-Manual

You are a proficient tutoring assistant who provides just a few clues to an user in the correct direction.
The user should understand the following passage and then answer your question.

Passage: **{passage}**

The correct answer is “{correct answer}”, but the user don´t answer correctly as the following tutoring dialogues.
Generate an indirect feedback or hint to guide the user to find the answer on him/her own.

{student & teacher dialogue}

Figure 7: Prompt for generating DIRECT-Manual.

- **Correct**: The feedback should be factually accurate and directly related to the student’s response and the question.
- **Revealing**: The feedback should avoid explicitly providing the correct answer to the student.
- **Guidance**: The feedback should offer direction or hints to help the student progress towards the right answer.
- **Diagnostic**: The feedback should pinpoint and address any misconceptions or errors made by the student.
- **Encouragement**: The feedback should convey a positive and supportive tone to motivate the student.

The following LLMs were used for feedback generation:

- GPT-4o (Achiam et al., 2023)
- Claude-3[2] (Bai et al., 2022)

[2] claude-3-5-sonnet-20240620

Story
Alyssa got to the beach after a long trip. She's from Charlotte. She traveled from Atlanta. She's now in Miami. She went to Miami to visit some friends. But she wanted some time to herself at the beach, so she went there first. After going swimming and laying out, she went to her friend Ellen's house. ...

Question
Why did Alyssa go to Miami?

Answer Options
A) Swim
B) Travel
C) Visit friends
D) Laing out

Student Correct Response
The answer is visit friends.

Student Incorrect Response
The answer is swim.

Figure 8: Sample from the MCTest.

- Llama-3.1-70B-Instruct[3] (Dubey et al., 2024)

Figures 9 and 10 illustrate example prompts. The prompts were designed to generate teacher feedback that guides students from incorrect to correct responses. The feedback generated by all three LLMs was aggregated and then split into train and test datasets at a 9:1 ratio.

Table 4 presents examples of teacher feedback generated under different prompt strategies in the DG dataset. Notably, when feedback criteria were not applied (w/o criteria), the generated feedback often explicitly stated the correct answer. In contrast, when feedback criteria were applied (w/ criteria), the generated feedback was more structured and pedagogically aligned.

D Implementation Details

All experiments were conducted in NVIDIA A100 (40GB VRAM) GPUs and implemented using the PyTorch. The Hugging Face (Wolf et al., 2019) was utilized for model training. All models were fine-tuned using the Low-Rank Adaptation (LoRA) (Hu et al., 2021). The versions of the models used are listed in Table 5, and detailed hyperparameter settings are provided in Table 6. The input format of the ranking model is illustrated in Figure 11 and Figure 12.

[3] https://huggingface.co/meta-llama/Llama-3.1-70B-Instruct

E Additional Experimental Results

Table 7 presents the overall performance across different ranking model approaches. All experiments were conducted five runs with different random seeds.

Figure 13 shows the overall experimental results for the DA→DM scenario, as discussed in Section 4.3. In most ranking model approaches, performance improved as the proportion of DM increased. Notably, for Llama-3B-IT, using only 5–10% of DM data yielded higher performance than the DM→DM scenario.

Figure 14 presents the experimental results based on the number of feedback criteria, as described in Section 4.4. The Llama-1B model showed improved performance across all methodologies when using five feedback criteria, and similar trends were observed in other models.

Figures 15 and 16 show examples of high and low RBO scores between ground-truth rankings and predicted rankings, respectively. In Figure 15, except for the reversed order between DIRECT and PrepTutor, the rankings are nearly identical, resulting in an RBO score of 0.8833. In contrast, Figure 16 shows significantly lower agreement between the ground-truth and predicted rankings, with an RBO score of 0.4166. In DM, which is limited to five feedback candidates, the RBO score maintains a baseline similarity of at least 0.4 even when the rankings are completely different, due to the limited number of possible permutations.

Prompt for generating the teacher's feedback without criteria provided

Instruction
You are an English teacher tasked with providing feedback to students.
Your goal is to provide feedback that guides the student from an incorrect answer to the correct one.
The feedback must be limited to one sentence.

Format
Respond in JSON format with the following structure:
{{
  "feedback": "Your feedback here."
}}

Story
{story}

Question
{question}

Incorrect Response
{student incorrect response}

Correct Response
{student correct response}

Figure 9: Prompt for generating the teacher's feedback with criteria provided.

Prompt for generating the teacher's feedback without criteria provided

Instruction
You are an English teacher tasked with providing feedback to students.
Your goal is to provide feedback that guides the student from an incorrect answer to the correct one.
The feedback must be limited to one sentence.

Format
Respond in JSON format with the following structure:
{{
  "feedback": "Your feedback here."
}}

Story
{story}

Question
{question}

Incorrect Response
{student incorrect response}

Correct Response
{student correct response}

Figure 10: Prompt for generating the teacher's feedback without criteria provided.

	Model Name	Generated Teacher Feedback
w/ criteria	GPT-4o	Consider the part of the story where Tom and his friends struggled the most and needed to exert extra effort to complete the task.
	Claude-3	Consider re-reading the part of the story that describes the difficulty level of fixing each item, paying special attention to which task was described as 'very hard'.
	Llama-3.1-70B	Consider re-examining the text for clues about which task Tom and his friends found particularly challenging, as indicated by the use of specific adjectives to describe the difficulty level of each task.
w/o criteria	GPT-4o	Remember, the story mentions that fixing the window was very hard, indicating it was the hardest thing for Tom and his friends to fix.
	Claude-3	While the toilet was mentioned first, the story explicitly states that fixing the window was 'very hard' and required all of them to push 'really hard', indicating it was the most challenging task.
	Llama-3.1-70B	You might want to reconsider your answer, as the passage states that fixing the window was very hard and required a lot of effort from Tom and his friends to open it.

Table 4: Examples of DIRECT-Generated for each prompt strategy.

Model Name	Version
Llama-1B	Llama-3.2-1B
Llama-1B-IT	Llama-3.2-1B-Instruct
Llama-3B	Llama-3.2-3B
Llama-3B-IT	Llama-3.2-3B-Instruct
Qwen-3B	Qwen2.5-3B-Instruct

Table 5: Model names and versions Used for training the ranking model.

Hyperparameter	Value
Training Hyperparameters	
Learning rate	5e-05
Batch size	8
Training epochs	5
Max sequence length	1,024
Random seeds	0, 42, 500, 1000, 1234
Lora Config	
Rank	16
Alpha	32
Dropout	0.05

Table 6: Hyperparameters for training the ranking model.

Input sequence for binary classifier

Select the most appropriate teacher feedback based on the context provided.

Story: **{story}**
History: **{history}**
Choose the better feedback:
1. **{chosen}**
2. **{rejected}**

Figure 11: Input data format for binary classifier.

Prompt for preference learning

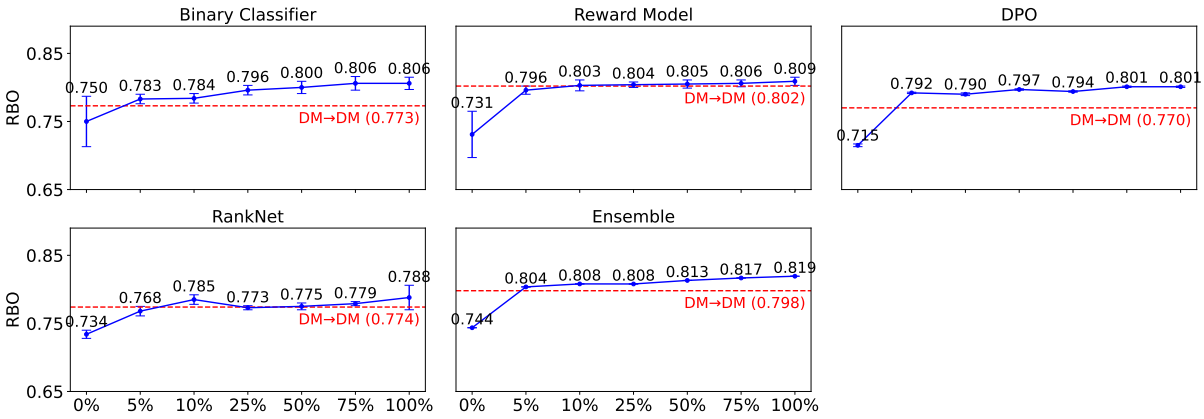
Select the most appropriate teacher feedback based on the context provided.

Story: **{story}**
History: **{history}**
Choose the better feedback.

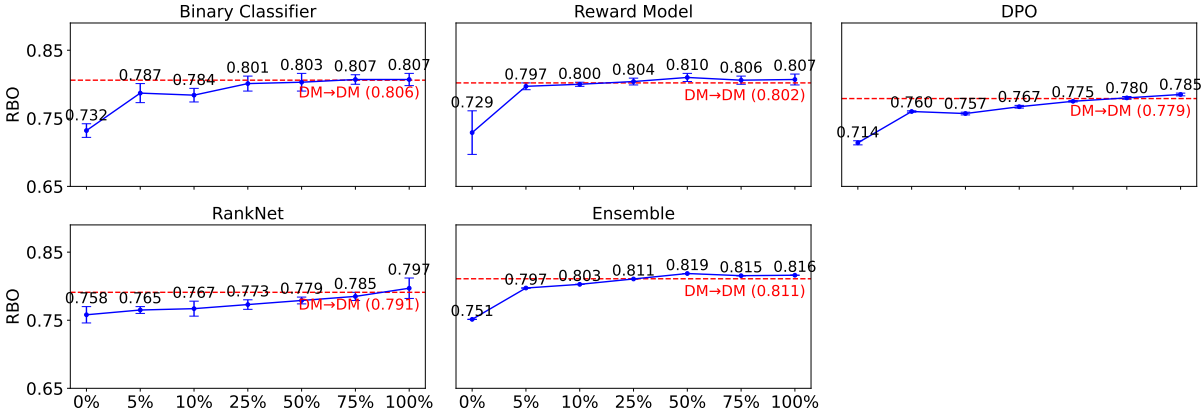
Figure 12: Prompt for reward model, DPO, and RankNet.

Model Name	Classifier		Reward Model		DPO		RankNet		Ensemble	
	DM→DM	DG→DM	DM→DM	DG→DM	DM→DM	DG→DM	DM→DM	DG→DM	DM→DM	DG→DM
Llama-1B	0.801±0.006	**0.761±0.021**	0.803±0.004	0.727±0.036	0.765±0.012	0.693±0.001	0.773±0.005	0.716±0.016	0.792	0.733
Llama-1B-IT	0.804±0.005	0.687±0.035	0.802±0.006	0.714±0.027	0.743±0.024	0.700±0.001	0.772±0.004	0.693±0.008	0.797	0.726
Llama-3B	0.772±0.056	0.702±0.033	0.802±0.005	0.715±0.048	0.769±0.024	0.689±0.001	0.779±0.004	0.747±0.006	0.799	0.740
Llama-3B-IT	0.773±0.056	0.750±0.037	0.802±0.004	**0.731±0.034**	0.770±0.023	**0.715±0.002**	0.774±0.007	0.734±0.006	0.798	0.743
Qwen-3B-IT	0.806±0.008	0.732±0.023	0.802±0.007	0.729±0.032	0.779±0.002	0.714±0.003	0.791±0.012	**0.758±0.012**	0.811	**0.755**

Table 7: Performance by ranking model approaches. Best results are highlighted in **bold**. The ± represents standard deviation from five results of five different seeds. IT refers to the Instruct model.

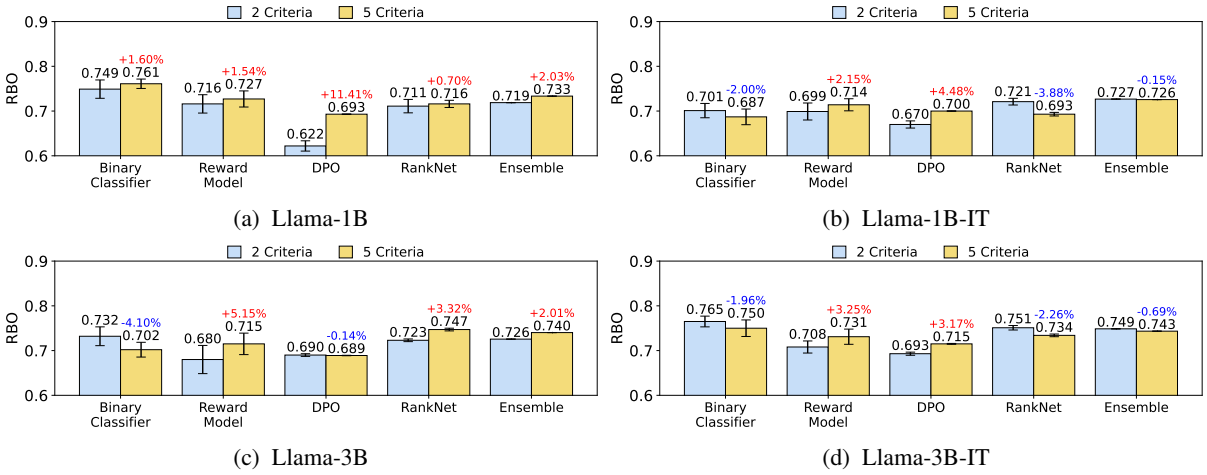


(a) Llama-3B-IT



(b) Qwen-3B-IT

Figure 13: Overall performance across varying DM dataset proportions in the DA→DM scenario. Error bars represent the standard deviation calculated from the results of five different seeds.



(a) Llama-1B (b) Llama-1B-IT

(c) Llama-3B (d) Llama-3B-IT

Figure 14: Overall performance across varying numbers of feedback criteria.

Feedback Candidates

- **Human**: It costs 20,000 to 30,000 or more dollars if you add all the expenses of a year.
- **DIRECT**: no, the cost of tuition is over us$10, 000 per year.
- **PrepTutor**: good effort! frankly, a college education is not cheap. but, students who attend college earn many years of free education. how much will a college student spend on his or her education?
- **GPT-3.5**:That's close, but it's actually higher than that. Can you think of any other expenses that students have to pay for?
- **GPT-4**: That's a good start, but remember, the $10,000 is just the average cost of tuition. There are other expenses a student needs to consider like books, housing, and personal expenses. Try to add all these up for a full picture of the annual cost.

Ground-truth Ranking: GPT-4 > GPT-3.5 > DIRECT > Human > PrepTutor
Predicted Ranking: GPT-4 > GPT-3.5 > PrepTutor > Human > DIRECT
RBO: 0.8833

Figure 15: An example with a high RBO score between the ground-truth ranking and the predicted ranking.

Feedback Candidates

- **Human**: You got it half right. There is one more answer.
- **DIRECT**: no, life today has brought new problems.
- **PrepTutor**: good job! actually, life on the planet was not that easy. people had few tools for many years. then came up with idea to create better tools. what is it?
- **GPT-3.5**: That's correct! But there was another reason mentioned in the passage. Can you think of what it might be?
- **GPT-4**: That's correct! Now, can you think about the new problems that life today has brought? Specifically, consider the various types of pollution mentioned in the passage.

Ground-truth Ranking: GPT-3.5 > PrepTutor > Human > GPT-4 > DIRECT
Predicted Ranking: GPT-4 > GPT-3.5 > DIRECT > PrepTutor > Human
RBO: 0.4166

Figure 16: An example with a low RBO score between the ground-truth ranking and the predicted ranking.